

Bayesian Learning and Monte Carlo Simulation

Airline Customer Satisfaction (REPORT SKELETON, NOT THE REPORT)

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This file is a scaffold, not a draft. Every section below says what has to be written, lists the numbers and figures needed for it, and suggests who is best placed to write it. The prose itself is deliberately not written. Replace each TODO block with your own text. Delete this box and the subtitle before submitting.
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1 Data and the problem

1.1 What this section has to do

The brief asks for a *brief description of the data and of the problem*. Two or three paragraphs, no more. It should answer: what are we predicting, from what, and why would anyone care.

1.2 TODO: write this

- State the prediction task and the response variable in one or two sentences. Say explicitly that it is binary, which is what forces the Bernoulli likelihood in Section 3.
- Write the likelihood and link equation once, here or in Section 3, not both. Decide which and be consistent.
- Describe the covariates by group rather than listing all 22: passenger and trip variables vs. the 14 service ratings.
- Describe the cleaning decisions and, importantly, *justify* each in one clause. Do not just say what was done.
- Say what the headline finding is, so the reader knows where the report is going. Most reports do this and it reads much better than a slow build-up.

Who: whoever presents this section.

1.3 Reference: facts you will need

Do not copy these sentences; they are here so you do not have to go digging.

Raw file	129,880 rows, 22 columns
Response	satisfaction , 54.7% satisfied, close to balanced
Missing data	393 rows (0.3%), arrival delay only; dropped, not imputed
Duplicates	none
Training set	$n = 1197$, stratified on satisfaction $\times$ cabin class
Held-out set	128,290 rows, never touched during fitting

Transformations that need justifying in the text, with the maths already written out:

Skewed covariates. Flight distance and departure delay are strongly right-skewed, so both enter as $\log(1 + x)$ before standardization.

Decorrelating the delay pair. Departure and arrival delay correlate at $r = 0.965$. Rather than including both, we regress arrival on departure delay and keep the residual,

$$\text{arr_delay_extra}_i = \text{arr_delay}_i - (\hat{\alpha} + \hat{\gamma} \text{dep_delay}_i),$$

which by the defining property of least squares is orthogonal to **dep_delay** (measured $r \approx 3 \times 10^{-15}$) and retains only the “time gained or lost in the air” signal.

Standardization. Every continuous covariate is centred and scaled using the **training** mean and standard deviation,

$$z_{ij} = \frac{x_{ij} - \bar{x}_j^{\text{train}}}{s_j^{\text{train}}},$$

applied unchanged to the held-out set. Worth one sentence in the text on why this matters (no information from the test set enters the fit).

2 Exploratory analysis

2.1 What this section has to do

Motivate the modelling choices that follow. Specifically, it has to make the reader agree that a shrinkage prior is needed before Section 3 introduces one.

2.2 TODO: write this

- Present the correlation structure of the 14 service ratings and name the clusters. **This section currently has no figure and needs one**, see the note below.
- Explain *why* correlated predictors are a problem for a flat-prior regression, in plain terms. This is the sentence that earns Section 3.
- Mention the structural zeros and say explicitly that we chose not to model them, and why. Examiners reward naming a limitation you decided against fixing more than they punish the limitation.
- Optional: one boxplot of a few ratings split by satisfaction.

Who: open. Whoever writes it should read `00_data_audit_eda_subsampling.Rmd` first.

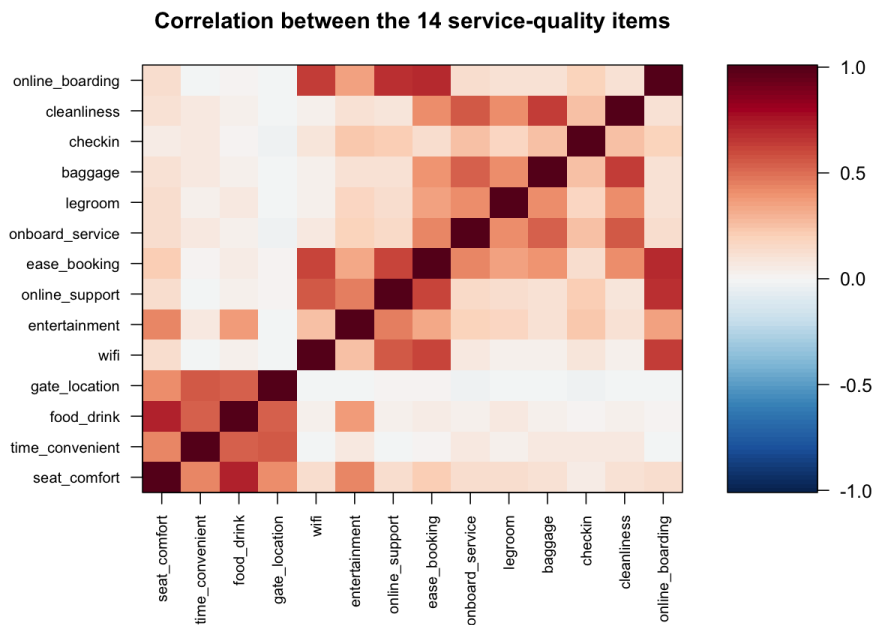


Figure 1: Correlation between the 14 service-quality items. The blocks along the diagonal are the clusters named below.

The correlation used here is the ordinary Pearson coefficient,

$$r_{jk} = \frac{\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_i (x_{ij} - \bar{x}_j)^2 \sum_i (x_{ik} - \bar{x}_k)^2}}.$$

2.3 Reference: the clusters

Cluster	Items	Pairwise r
Digital	wifi, online support, ease of booking, online boarding	0.56-0.63
Onboard	onboard service, legroom, baggage, cleanliness	0.4-0.6
Boarding logistics	seat comfort, time convenient, food and drink, gate location	n/a

Structural zeros: time convenient 5.1%, food and drink 4.6%, seat comfort 3.7%, entertainment 2.3%. Everything else is essentially 0%.

3 Model specification

3.1 What this section has to do

The brief is specific here: *describe the model for the data (likelihood) and the prior distribution, be clear about the parametrization and the distributions used*, and discuss the choice of hyperparameter values. This is the section most likely to be probed in the oral.

3.2 TODO: write this

- Write out the likelihood once, properly, with the link.
- Write out both priors: the flat $\mathcal{N}(0, 100)$ and the horseshoe. For the horseshoe, explain *what the two scales do*: global τ sets overall sparsity, local λ_j lets one coefficient escape it. Do not only state the distributions.
- Explain the shrinkage weight $\kappa_j = 1/(1 + \lambda_j^2)$ and what its two extremes mean. Section 4 leans on this, so define it here.
- Justify the hyperparameters. Why variance 100 and not 10 or 1000? The honest answer is that Section 4.6 shows it does not matter, so say that and point forward.
- **Mention the precision-vs-variance point.** JAGS's `dnorm` takes precision, so variance 100 is written `dnorm(0, 0.01)`. It is a one-line remark and exactly the kind of implementation detail the examiner asks about.
- Include the MCMC settings table and say why the horseshoe needs a longer run than the flat model.

Who: Matteo wrote 03, so the horseshoe half is naturally his.

3.3 Reference: the model as implemented

These are the equations as actually coded, ready to drop into your text. The maths is fixed; the surrounding explanation is what you have to write.

Likelihood (both models):

$$y_i \mid p_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum_{j=1}^K X_{ij}\beta_j, \quad i = 1, \dots, n.$$

Because the response is binary there is no residual variance to estimate: the dispersion is fully determined by p_i .

Flat prior ($K = 9$):

$$\beta_0 \sim \mathcal{N}(0, 100), \quad \beta_j \stackrel{iid}{\sim} \mathcal{N}(0, 100), \quad j = 1, \dots, K.$$

JAGS parametrizes the normal by **precision** $\tau = 1/\sigma^2$, so a variance of 100 is coded `dnorm(0, 0.01)`.

Horseshoe prior ($K = 22$), Carvalho, Polson and Scott (2010):

$$\beta_j \mid \lambda_j, \tau \sim \mathcal{N}\left(0, \lambda_j^2 \tau^2\right), \quad \lambda_j \stackrel{iid}{\sim} \mathcal{C}^+(0, 1), \quad \tau \sim \mathcal{C}^+(0, 1),$$

with $\beta_0 \sim \mathcal{N}(0, 100)$ as before. Here $\mathcal{C}^+(0, 1)$ is the standard half-Cauchy, i.e. a Student- t with 1 degree of freedom truncated to the positive line, coded in JAGS as `dt(0, 1, 1) T(0,)`.

Shrinkage weight. Reparametrizing the local scale gives

$$\kappa_j = \frac{1}{1 + \lambda_j^2} \in [0, 1],$$

so that $\kappa_j \rightarrow 0$ means the coefficient is left essentially unshrunk (signal) and $\kappa_j \rightarrow 1$ means it is pulled to zero (noise). Under the prior, $\kappa_j \sim \text{Beta}(1/2, 1/2)$, the U-shaped density the “horseshoe” is named after.

Posterior predictive probability for a held-out passenger i , over S posterior draws:

$$\hat{p}_i = \frac{1}{S} \sum_{s=1}^S \text{logit}^{-1}\left(\beta_0^{(s)} + \sum_{j=1}^K X_{ij}\beta_j^{(s)}\right).$$

Note this is the average of the transform, not the transform of the average: $\mathbb{E}[g(\theta)] \neq g(\mathbb{E}[\theta])$ for nonlinear g .

Predictor sets. Flat model, 9 predictors: age, seat comfort, log flight distance, cabin class (2 dummies, Eco as reference), log departure delay, arrival-delay residual, travel type, customer type. Horseshoe model, 22 predictors: those 9 plus the 13 remaining service ratings.

MCMC settings actually used. **Verify against 02 and 03 before submitting**, these go stale if anyone retunes the samplers:

	Flat	Horseshoe
Chains	4	4
Adaptation	1,000	1,000
Burn-in	2,000	5,000
Saved per chain	2,000	10,000
Thinning	1	10
Total draws	8,000	4,000

4 Posterior analysis and model comparison

4.1 What this section has to do

The largest section. The brief wants posterior plots, summary statistics (means, variances, credible intervals), a sensitivity analysis, and an explicit statement of which algorithm was used. Model selection also lives here.

4.2 TODO: convergence

Write a short paragraph reporting the actual diagnostics. You need:

- maximum $\hat{R}$ across all parameters, from 02 and 03
- minimum effective sample size for the β_j 's
- a sentence saying the λ_j 's mix more slowly than the β_j 's, why that is expected, and why it is acceptable

Trace plots go in the appendix, not here. Say that explicitly so the reader knows where to find them.

4.3 TODO: interpreting the coefficients

- Say which covariates have credible intervals excluding zero, and in which direction.
- **Convert at least two or three coefficients to odds ratios** (e^{β_j}). Log-odds are hard to read; odds ratios are not. The reference report does this well and it is an easy way to look competent.
- Comment on the ones that are *not* distinguishable from zero. Eco Plus vs Eco and the arrival-delay residual are both interesting negatives.

4.4 TODO: reading the shrinkage figure

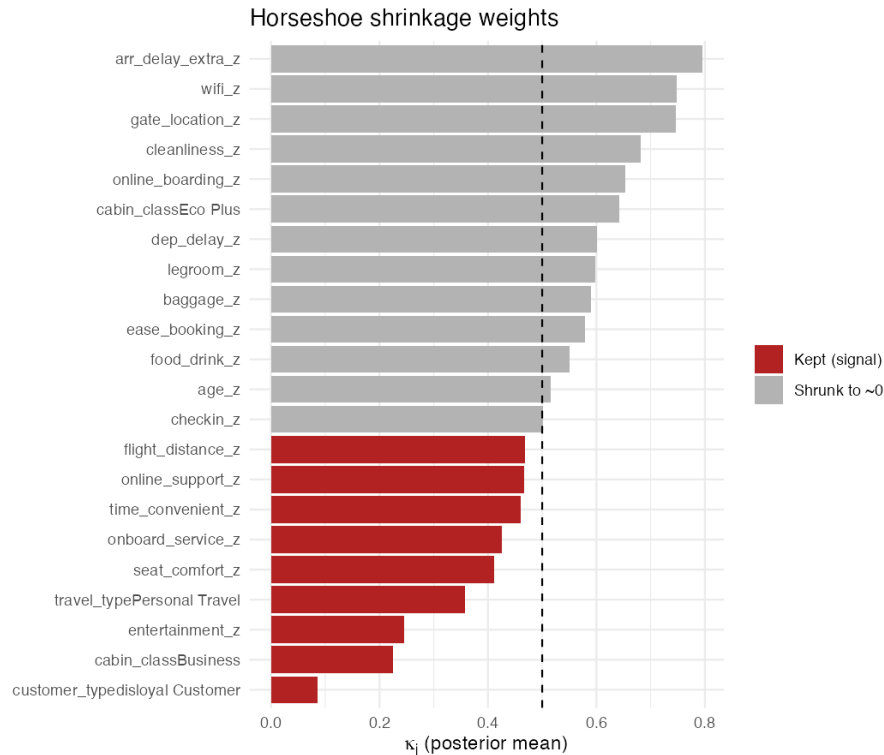


Figure 2: Horseshoe shrinkage weights κ_j . Low = kept as signal, high = shrunk toward zero.

Write 3-4 sentences on this. Cover:

- which predictors clearly survive and which are clearly shrunk
- the fact that most predictors sit in a middle band around 0.4-0.7 rather than splitting cleanly in two. **Say this honestly.** The sharp bimodal shape people associate with the horseshoe describes the *prior*, not a promise about the posterior. With $n \approx 1200$ the data are not sharp enough to decide every coefficient, and that is a finding, not a failure.

4.5 TODO: reading the flat-vs-horseshoe figure

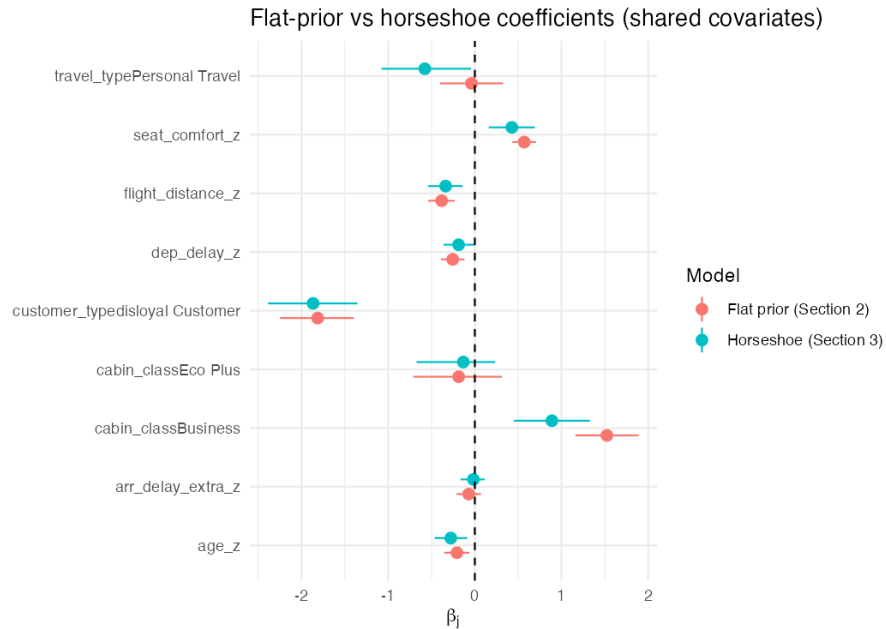


Figure 3: Posterior means and 95% credible intervals under the flat prior and the horseshoe, for the predictors both models share.

Write 3-4 sentences. Points worth making:

- Business class shrinks from about 1.5 to 0.85 but stays clearly away from zero: a real effect, resized once it competes with 13 more predictors.
- Customer type barely moves.
- Eco Plus and the arrival-delay residual were never distinguishable from zero under either prior; the horseshoe only confirms it.

4.6 TODO: the BAS cross-check

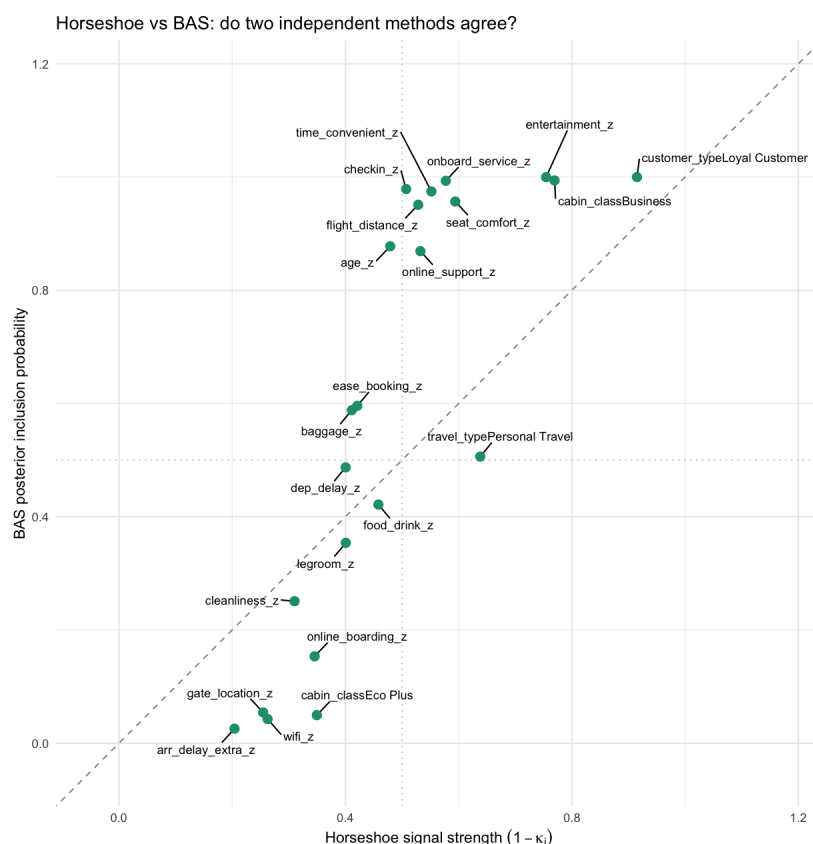


Figure 4: Horseshoe shrinkage weights against BAS posterior inclusion probabilities.

- Explain what `bas.glm` does differently: model averaging over a discrete model space rather than continuous shrinkage. Say why MCMC over models was used instead of full enumeration (2^{22} models).
- Say how well the two agree, and **name any predictor where they disagree**. Disagreement is the interesting part and shows you actually looked.

Who: Matteo wrote 04.

4.7 TODO: WAIC and PSIS-LOO

Present the comparison and then add the caveat, which matters:

This is **not** a clean test of “horseshoe prior vs. flat prior”. The flat model does not just have a different prior, it has 13 fewer predictors. The honest claim is that the larger model with shrinkage beats the smaller model with flat priors, and the two causes cannot be fully separated.

Write that caveat in your own words. It is the kind of self-awareness that gets rewarded in the oral.

Also present the two within-model comparisons and interpret them:

- dummy vs ordinal `cabin_class`: connect the WAIC result back to the coefficients (Business jumps, Eco Plus does not, so equal spacing is wrong)

- logit vs probit: they are indistinguishable, so say the choice was made on interpretability, not fit

4.8 TODO: DIC

The change-point lecture notebook introduces DIC alongside WAIC, so reporting both is closer to what the course actually teaches than WAIC alone.

Write a short paragraph that:

- states the DIC values and the effective parameter counts p_D for both models (numbers are in 05, table under the “DIC” heading)
- says whether DIC ranks the models the same way WAIC does. If it does, say so, two criteria agreeing is a stronger claim than one. If it does not, explain why they can differ: DIC penalizes using a single point estimate $D(\bar{\theta})$, WAIC uses the whole posterior predictive density, and the horseshoe’s effective parameter count is exactly the sort of quantity the two treat differently
- mentions that DIC comes straight out of `dic.samples()` in JAGS while WAIC needs the pointwise log-likelihood monitored inside the model

Definition, if you want it in the text:

$$\text{DIC} = \overline{D(\theta)} + p_D, \quad p_D = \overline{D(\theta)} - D(\bar{\theta}), \quad D(\theta) = -2 \log f(y \mid \theta),$$

where $\overline{D(\theta)}$ is the posterior mean deviance and $\bar{\theta}$ a posterior point estimate. Smaller DIC is better.

4.9 TODO: prior sensitivity

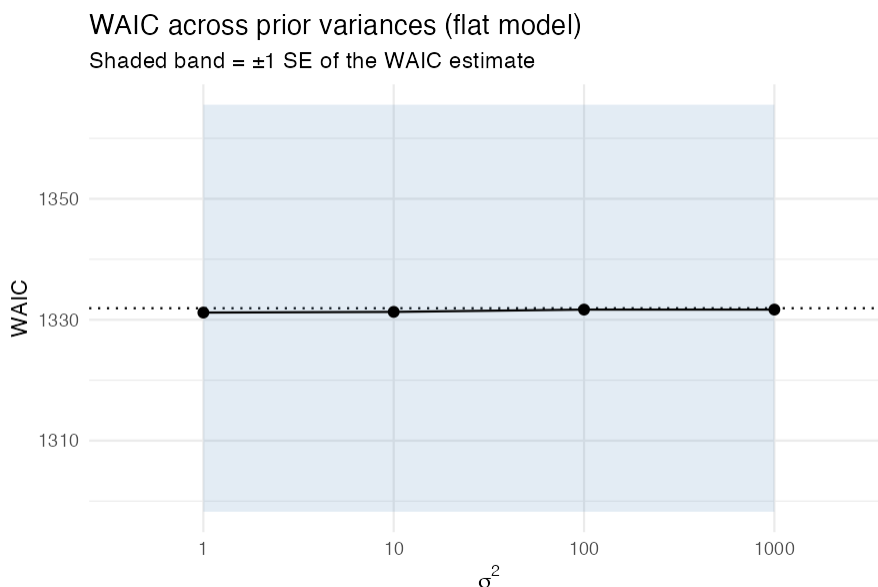


Figure 5: WAIC across prior variances for the flat model. Shaded band is ± 1 SE.

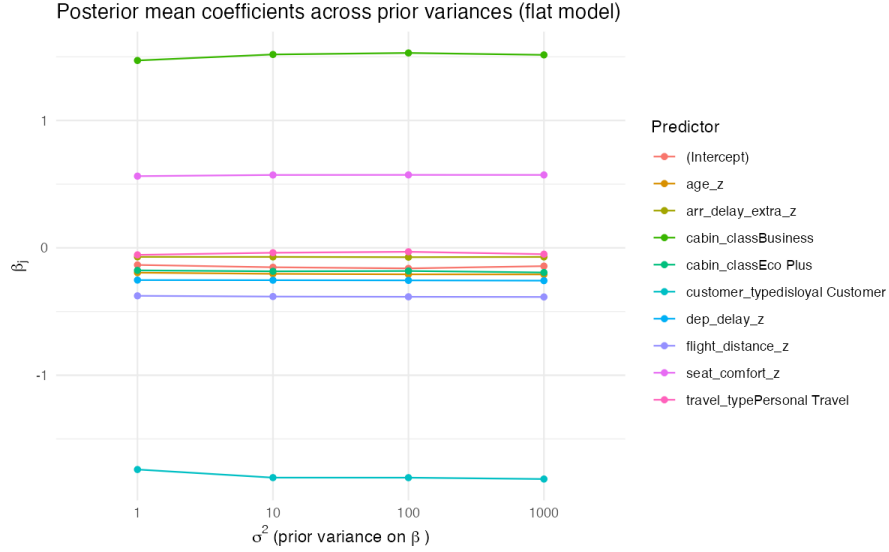


Figure 6: Posterior mean coefficients across prior variances.

- Report the WAIC spread against the WAIC standard error. The comparison is the point: a spread of 0.67 against an SE of 33.7 means the prior barely matters.
- Conclude that $\sigma^2 = 100$ is a safe default rather than an arbitrary choice.
- **Also mention the conjugate baseline from 01:** three different Beta priors agree to three decimal places at $n \approx 1200$. It is a second, cleaner illustration of the same point and costs two sentences.

4.10 Reference: definitions, if you want them in the text

WAIC, as computed by `loo` from the pointwise log-likelihood monitored inside JAGS:

$$\text{WAIC} = -2 \left(\widehat{\text{elpd}} - p_{\text{waic}} \right), \quad \widehat{\text{elpd}} = \sum_{i=1}^n \ln \left(\frac{1}{S} \sum_{s=1}^S p(y_i | \theta^{(s)}) \right),$$

with $p_{\text{waic}} = \sum_i \text{Var}_s \left[\ln p(y_i | \theta^{(s)}) \right]$ the effective number of parameters. Lower WAIC is better.

4.11 Reference: all the numbers for this section

Model comparison:

Model	ELPD	p_{eff}	WAIC	SE
Flat (9 predictors)	-665.8	10.1	1331.6	33.7
Horseshoe (22 predictors)	-490.0	24.7	979.9	40.7

`loo_compare`: flat is 175.8 ELPD points behind the horseshoe, SE 16.7.

DIC (from 05), with the WAIC values repeated for comparison:

Model	Deviance	p_D	DIC	WAIC
Flat (9 predictors)	1321.2	9.8	1331.0	1331.6
Horseshoe (22 predictors)	954.2	22.9	977.1	979.9

Two things worth using in the text. The two criteria agree almost exactly and rank the models the same way, which is a stronger claim than either alone. And p_D behaves as it should: the flat model has 10 parameters and p_D comes out at 9.8, while the horseshoe has 46 nominal parameters (22 slopes, 22 local scales, τ , intercept) but an effective count of only 22.9, so the shrinkage really is buying back complexity.

Within the flat model:

Comparison		WAIC	SE
cabin_class	dummy	1327.7	33.8
	ordinal	1340.1	33.1
Link	logit	1327.7	33.8
	probit	1329.0	33.4

Flat-model posterior means and 95% intervals (standardized scale):

Covariate	Mean	95% CI	Excludes 0?
intercept	-0.147	(-0.469, 0.172)	no
age	-0.209	(-0.359, -0.063)	yes
log flight distance	-0.369	(-0.528, -0.215)	yes
seat comfort	0.568	(0.429, 0.706)	yes
departure delay	-0.314	(-0.464, -0.161)	yes
arrival-delay residual	-0.046	(-0.182, 0.094)	no
cabin: Eco Plus	-0.187	(-0.682, 0.319)	no
cabin: Business	1.504	(1.148, 1.872)	yes
travel: Personal	-0.042	(-0.405, 0.341)	no

Customer type is missing from that table, pull it from 02 output.

Horseshoe shrinkage weights, least to most shrunk: customer type disloyal 0.09, Business 0.24, entertainment 0.25, travel personal 0.37, seat comfort 0.42, onboard service 0.44, then a middle band 0.46-0.69, then gate location 0.74, wifi 0.74, arrival-delay residual 0.80.

Prior sensitivity: WAIC ranges 1331.26-1331.93 across $\sigma^2 \in \{1, 10, 100, 1000\}$, a spread of 0.67. Largest coefficient shift 0.089 (customer type).

5 Prediction on held-out data

5.1 What this section has to do

The brief calls a prediction exercise *required* for this dataset. Fit on one part, predict the other, and report honestly.

5.2 TODO: write this

- State the size of the held-out set and stress that it was never touched during fitting, standardization or variable selection.
- **Explain how the predicted probability is computed.** It is the posterior mean of $\text{logit}^{-1}(\beta_0 + x_i^\top \beta)$ averaged over draws, *not* logit^{-1} evaluated at the posterior mean of β . The two differ because the link is nonlinear. This is a good detail to include and a very likely oral question.
- Present the results table.
- Explain what each metric measures and why all three are reported: accuracy is one arbitrary threshold, AUC is threshold-free ranking, calibration is whether the probabilities mean anything.
- **The calibration paragraph is the distinctly Bayesian one.** Make the point that a model can rank perfectly and still be systematically overconfident, and that for a Bayesian model that would be a real failure.
- Comment on the confusion matrices: whether errors lean toward false positives or negatives. The response is close to balanced so a lopsided pattern would be worth flagging.
- Draw the headline conclusion: horseshoe and BAS land in essentially the same place by completely different mechanisms. Two independent methods agreeing is stronger than either winning alone.

5.3 TODO: figures

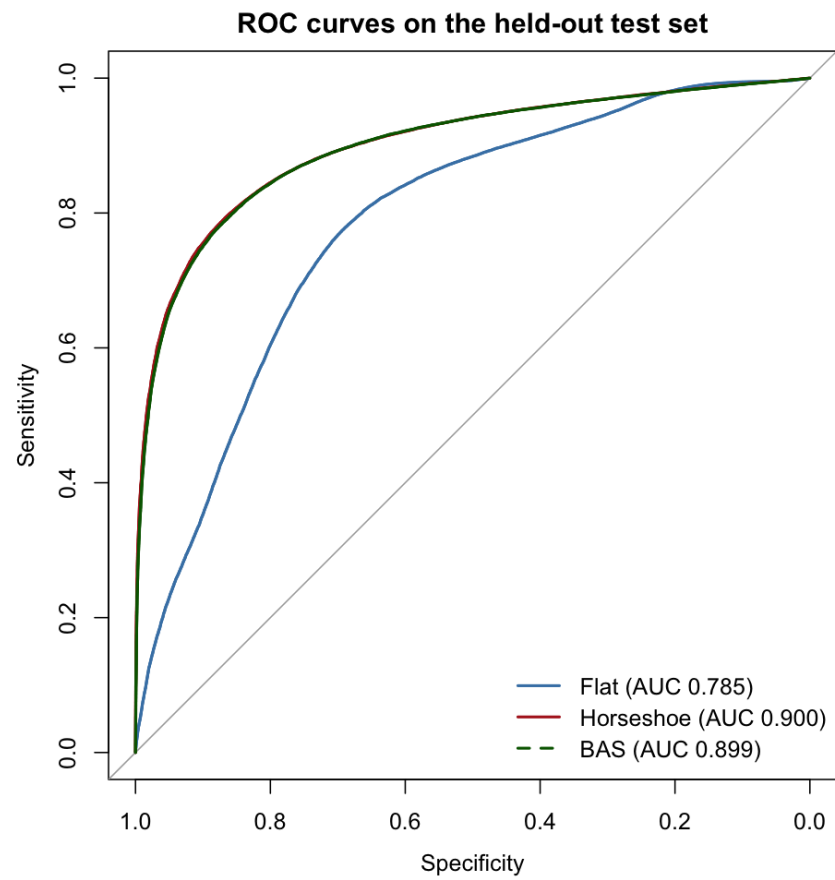


Figure 7: ROC curves on the held-out test set.

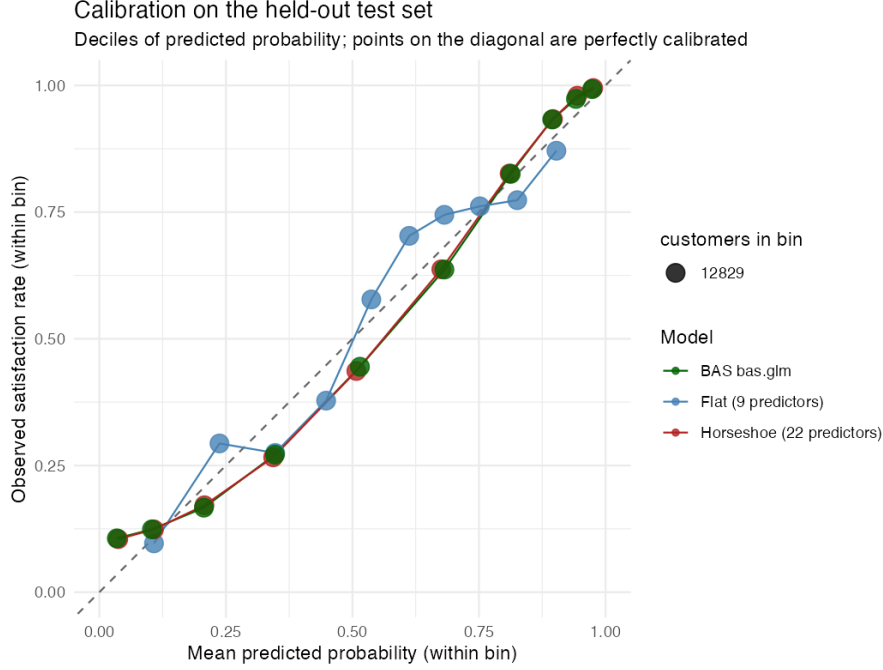


Figure 8: Calibration on the held-out test set. Points on the diagonal are perfectly calibrated.

5.4 TODO: optional, but would score well

The reference report turns its predictions into a business recommendation. We could do the same: given that customer type and cabin class dominate, which passengers should the airline prioritize? Half a page, and it makes the conclusions concrete rather than abstract.

5.5 Reference: definitions and numbers

Expected calibration error, over B equal-count bins of predicted probability:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{n} |\bar{p}_b - \bar{y}_b|,$$

where n_b is the number of passengers in bin b , $\bar{p}_b$ their mean predicted probability and $\bar{y}_b$ the observed satisfaction rate in that bin. Zero is perfect calibration.

Results, held-out set of 128,290 passengers, accuracy at a 0.5 cutoff:

Model	Accuracy	AUC	ECE
Flat (9 predictors)	73.9%	0.785	0.050
Horseshoe (22 predictors)	82.4%	0.900	0.041
BAS bas.glm	82.3%	0.899	0.042

Confusion matrices are printed in 06.

6 Conclusions

6.1 What this section has to do

Half a page. Pull the threads together, do not introduce anything new.

6.2 TODO: write this

Cover, in whatever order reads best:

- what actually drives satisfaction: customer loyalty and cabin class dominate everything; entertainment is the strongest service item once all 14 compete; wifi and the arrival-delay residual turn out redundant
- that the conclusions are prior-insensitive
- that shrinkage and model averaging, two independent methods, agree
- the limitations we chose not to fix:
 - structural zeros in four service items are not modelled
 - encoding and link were chosen on training WAIC, not re-validated out of sample
 - $n \approx 1197$ is small relative to 22 predictors

Who: whoever presents the conclusions should write them, in their own words. It is the section most likely to be quoted back at you.

Appendix

A. MCMC diagnostics

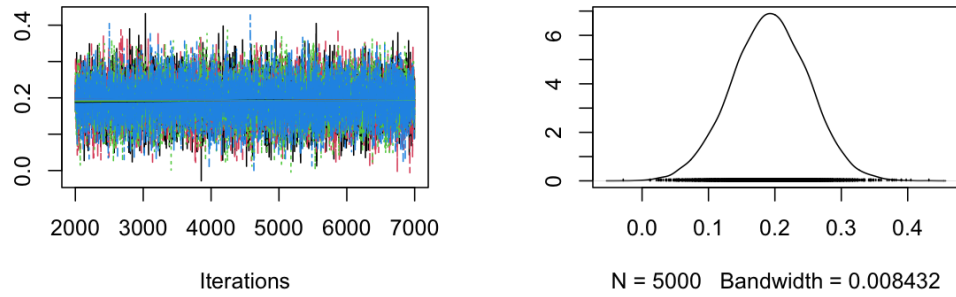


Figure 9: Trace and posterior density for the intercept β_0 in the intercept-only model. Four chains overlaid.

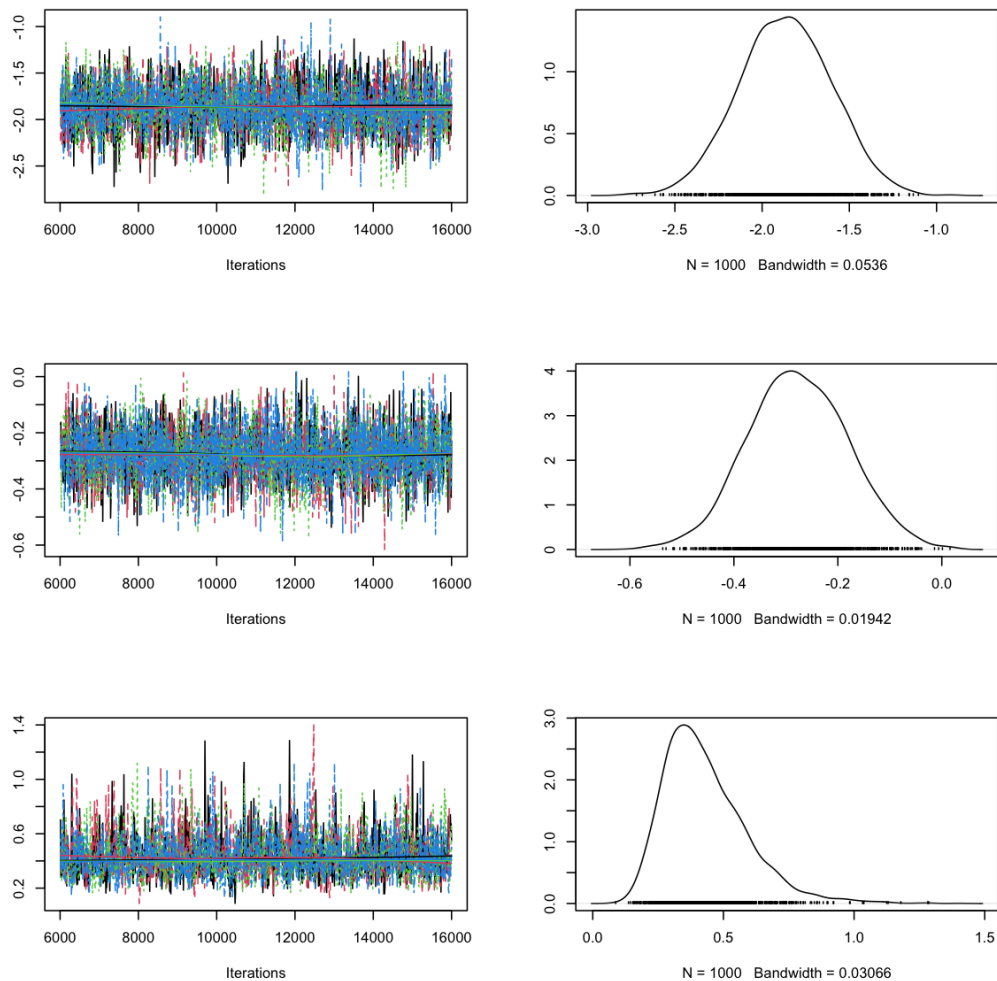


Figure 10: Horseshoe model. Top: the least shrunk coefficient. Middle: a heavily shrunk one. Bottom: the global scale τ .

TODO: write the accompanying paragraph. Two or three sentences reading these plots: chains overlapping with no drift or trend, densities smooth and unimodal, and the observation that τ mixes visibly worse than the β_j 's, which is expected under heavy-tailed hyperpriors.

TODO: add the numerical diagnostics. Paste in the Gelman-Rubin table and the effective sample sizes from 02 and 03. Definitions, if you want them in the text:

$$\hat{R} = \sqrt{\frac{\widehat{\text{Var}}(\theta)}{W}}, \quad N_{\text{eff}} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k},$$

where W is the within-chain variance and ρ_k the autocorrelation at lag k . Values of $\hat{R}$ near 1 indicate the chains have converged to the same distribution.

B. Pareto k diagnostics

TODO. 05 produces these for the PSIS-LOO comparison. One short paragraph confirming the comparison can be trusted at face value.

C. Reproducing the analysis

TODO. Short. The pipeline is seven notebooks, 00 through 06, knitted in order; setup and a JAGS version caveat are in `README.md`.

Open question for the professor: the brief asks for “separate R file(s), ready to be run”. We have `.Rmd` notebooks, not plain `.R` scripts. Someone should ask whether that is acceptable, or whether we should also ship `.R` versions via `knitr::purl()`.

Checklist before submitting

- ☐ Delete the box on page 1 and the “REPORT SKELETON” subtitle
- ☐ Fill in the three student IDs
- ☐ Every TODO block replaced with real text
- ☐ Body is 8-10 pages, appendix on top of that
- ☐ **No code anywhere in the report**, it is submitted separately
- ☐ Correlation heatmap saved from 00 and included in Section 2
- ☐ Trace plots saved and included in Appendix A
- ☐ MCMC settings table re-checked against 02 and 03
- ☐ Customer-type row added to the coefficient table
- ☐ Confirmed with the professor whether `.Rmd` is acceptable as the code submission
- ☐ Submitted at least 3 days before the exam date